AgriGuardian

Intelligent Agriculture. Empowered Farmers.

SOLUTION PITCH

An AI-powered farming companion that turns scattered data into **clear, verified decisions** — by voice, in 10 Indian languages, even offline.

THE PROBLEM

Farmers decide on information that is scattered, late and uncertain.

- Fragmented agricultural information
- Crop diseases, diagnosed too late
- Unpredictable weather and climate risk
- Uncertain market prices
- Limited access to agricultural experts
- Poor financial visibility and connectivity

OUR SOLUTION

One integrated, AI-powered agricultural intelligence platform that supports farmers through the entire farming lifecycle.

Sow → Grow → Protect → Harvest → Sell

- Voice-first, in 10 Indian languages
- Grounded in cited sources, checked by experts
- Built to work with poor connectivity

KEY INNOVATIONS

Gnani AI Voice Assistant

Speak in 10 Indian languages and hear spoken answers. Transcripts can be reviewed and corrected.

AI Crop Disease Detection

Leaf-photo diagnosis with Grad-CAM heatmaps and severity staging. Tomato model today.

Evidence-Based Advisory

Recommendations grounded in cited ICAR sources, with guardrails against unsafe chemical advice.

Weather & Market Intelligence

Farm-level forecasts and crop-risk alerts, plus mandi prices and net-realization estimates.

Expert Verification

Low-confidence or severe cases escalate to agronomists; farmers see a verification badge.

Offline-First PWA

Installable and usable without signal; queued changes sync safely when connectivity returns.

Farm Economics

Track expenses, harvests and sales to see cost per quintal, break-even and profit.

BUILT ON

- Gnani AI speech models
- PyTorch vision model
- Open-Meteo weather data
- ICAR-sourced knowledge

HOW IT WORKS

Farmer Input → Voice question, leaf photo and farm context

AI Intelligence → Vision, weather, market and knowledge engines

Verified Insights → ICAR-grounded evidence and expert review

Actionable Recommendations → Clear next steps in the farmer's language

Better Farm Decisions → Timely, informed choices from sowing to sale

INTENDED IMPACT

- **Earlier awareness of crop-health risks**
- **More informed agricultural decisions**
- **Better access to expert guidance**
- **Improved financial visibility**
- **Greater accessibility in low-connectivity areas**

From the first seed to the final sale — AgriGuardian connects every farming decision.